

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. USER PERSONA DEFINITION

- **Target User:** Content creators, amateur writers, and hobbyist storytellers.
- **Primary Goal:** To seamlessly transform an original story idea into a fully formatted, multi-panel visual comic without interacting with complex graphic design software.

3. JOURNEY STAGES & TOUCHPOINTS

Phase 1: Discovery & Interface Interaction

- **User Action:** The user launches the ComicCraft application in their web browser.
- **Touchpoint:** The frontend web interface (`index.html`).
- **User Emotion:** Curious about the AI's capabilities but cautious about the technical difficulty.
- **System Opportunity:** Provide a highly intuitive, form-based layout with clear dropdown menus for selecting art styles and tones, ensuring a frictionless onboarding experience.

Phase 2: Parameter Configuration & Submission

- **User Action:** The user inputs a specific narrative prompt, names the protagonist, selects a genre, and clicks the generation button.
- **Touchpoint:** The FastAPI POST route.
- **User Emotion:** Anticipation and excitement.
- **System Opportunity:** Implement visual feedback (e.g., a loading animation) to assure the user that the request is actively being processed by the backend servers.

Phase 3: Background AI Processing (Invisible to User)

- **System Action:** The server asynchronously communicates with Google Gemini to structure the narrative and Hugging Face Stable Diffusion to synthesize the visual assets.
- **Touchpoint:** External Cloud APIs.
- **System Goal:** Handle external network requests efficiently and deploy fallback mechanisms instantly if API rate limits are reached, preventing the application from crashing.

Phase 4: Visual Feedback & Review

- **User Action:** The user views the generated comic story sequentially on their screen.
- **Touchpoint:** The dynamic Jinja2 rendering template (`preview.html`).
- **User Emotion:** Amazement at seeing their written words transformed into visual art.
- **System Opportunity:** Ensure the web layout cleanly separates the generated artwork from the AI-written dialogue for maximum readability.

Phase 5: Document Retrieval

- **User Action:** The user clicks the download link to save their comic book.
- **Touchpoint:** File download endpoint.
- **User Emotion:** High satisfaction and readiness to share the creation.
- **System Opportunity:** Utilize the FPDF module to deliver a perfectly aligned, offline-ready PDF document.